



Evaluation of Decay Functions for Vector Field Based Obstacle Avoidance

Yonry Zhu*, Travis Moleski† and Jay P. Wilhelm‡
Mechanical Engineering, Ohio University, Athens, Ohio, 45701

Obstacle avoidance may be incorporated into vector field guidance using repulsive vector fields. The influence of these repulsive vector fields is limited to a certain region by scaling the field strength with a decay function. The present work examines the effects of five different obstacle vector field decay functions on vehicle performance using gradient vector field guidance. Three scenarios were simulated: single object avoidance with different vehicle velocities, single object avoidance with different vehicle approach angles, and multiple object avoidance. For the examined conditions, the best performance was achieved using a hyperbolic tangent decay function when the vehicle velocity to obstacle radius ratio is less than 10. It appears that effective avoidance is achieved when the obstacle field strength rapidly drops to zero once outside of the obstacle radius.

I. Nomenclature

x_c	=	Center X Component of Obstacle
y_c	=	Center Y Component of Obstacle
$\vec{V}$	=	Resultant GVF
G	=	Coefficient of Convergence Field
V	=	Potential Function
H	=	Coefficient of Circulation Field
α_i	=	Surface Function
L	=	Coefficient of Time Varying Field
M	=	Time Varying Expression
$\vec{V}_{conv}$	=	Convergence Field
$\vec{V}_{circ}$	=	Circulation Field
$\vec{V}_{TV}$	=	Time Varying Field
θ	=	UAV Heading Angle
$\dot{x}$	=	Velocity in X Direction
$\dot{y}$	=	Velocity in Y Direction
$\dot{\theta}$	=	Heading Rate
x	=	X Position
y	=	Y Position
z	=	Z Position
r_{OVF}	=	Radius of Obstacle Field
$\bar{x}$	=	X Distance from Center of Obstacle
$\bar{y}$	=	Y Distance from Center of Obstacle
r	=	OVF Radius
R_A	=	Percentage of the OVF radius at which the field strength is A
θ_a	=	Vehicle Approach Angle
v	=	Vehicle Velocity
$\tilde{r}$	=	Nondimensionalized Radius
P	=	Decay Function

*Undergraduate Student, Mechanical Engineering, Ohio University, Athens, Ohio, 45701, Student Member

†Graduate Student, Mechanical Engineering, Ohio University, Athens, Ohio, 45701, Student Member

‡Assistant Professor, Mechanical Engineering, Ohio University, Athens, Ohio, 45701, Senior Member

θ_0 = Initial Vehicle Approach Angle
 δ = Angle of Departure

II. Introduction

Vector Fields (VF) can provide guidance for Unmanned Aerial Vehicles (UAVs), that generate path intercept and following direction. There are several different types of VF guidance methods, where some focus on generating a path and others direction towards a path. Potential field methods have been the most common place and are able to generate a path based on other objects within the field. In contrast, Gradient Vector Field (GVF) methods generate a field of guidance towards a predetermined path. The capability to generate guidance that follows a path that may include tight area and mission constraints and can also react to obstacle without a path re-plan is highly desired. Typically, smaller Tier I UAVs lack the capabilities to re-plan missions, due mostly to computational limitations, but may be able to receive new information about areas to avoid. The GVF can generate guidance data from a path and include new obstacle information, but the influence of an obstacle on a resultant calculated vector field guidance command needs to be represented. There have been several investigations [1–7] into obstacle radius decay representation and the influence on vehicle guidance, but the application to GVF and driving parameters of the decay functions have not yet been determined. This paper will focus on representing several different decay functions and their influence on obstacle avoidance for a UAV being guided by a GVF. Simulations will be ran using a Dubins vehicle model and experiments will be ran using a CrazyFlie 2.0 with turn rate constraints to emulate fixed wing dynamics.

III. Literature Review

Potential field methods, which utilize gradient descent, may encounter issues with local minima traps and may not be able to produce paths through closely spaced obstacles [8]. Potential field methods are also poorly suited for certain scenarios of interest in UAV guidance such as circumnavigation of a moving target. Path following VFs may be generated using a variety of methods. Two of the most prominent are the Lyapunov VF method described in [9] and the GVF method described in [10]. The fields generated by these two methods bear a topical resemblance, but their governing equations are completely different. The present work utilizes the GVF method, which is capable of producing time-varying VFs of higher complexity than the Lyapunov method. Briefly, the GVF method generates convergence, circulation, and time-varying VF components at the intersection of several surfaces.

The total VF can be expressed as

$$\vec{V} = G\nabla V + H \wedge_{i=1}^{n-1} \nabla \alpha_i - LM(\alpha)^{-1} a(\alpha). \quad (1)$$

For two intersecting surfaces described by α_1 and α_2 , the potential function V is given by

$$V = -\sqrt{\alpha_1^2 + \alpha_2^2}. \quad (2)$$

G , H , and L are coefficients that change the relative weighting of each field component. The total VF in component form is given as

$$\vec{V} = G\vec{V}_{conv} + H\vec{V}_{circ} + L\vec{V}_{TV}. \quad (3)$$

Sample vector fields obtained using this method for circumnavigation and straight line following are shown in Fig. 1 and Fig. 2 respectively. Obstacle avoidance may be implemented with the GVF method by creating a circumnavigation field with a negative weighting coefficient of the convergence term. This, in conjunction with a path radius much smaller than the desired obstacle radius, will ensure a repulsive field. These repulsive Obstacle Vector Fields (OVFs) may then be added to the total guidance field to form a total guidance vector.

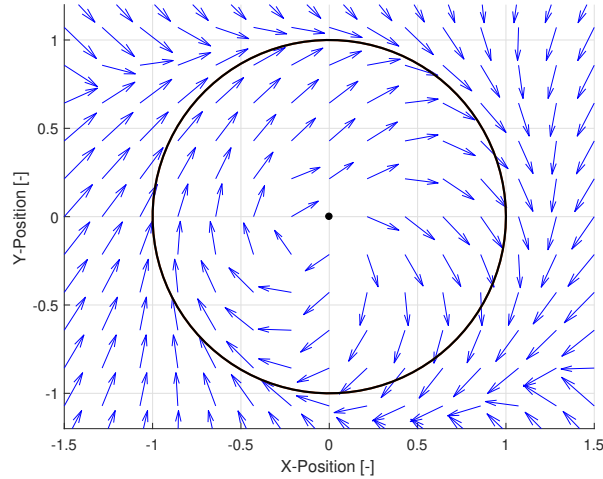


Fig. 1 Circumnavigation vector field generated by the intersection of a cylinder and plane using the Gradient Vector Field method. The black line shows the desired path. The black dot is the center of the vector field, $[x_c, y_c]$.

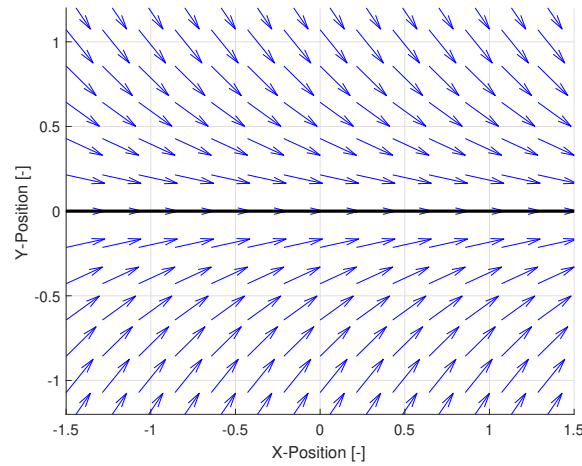


Fig. 2 Straight line following vector field generated by the intersection of two planes using the Gradient Vector Field method. The black line indicates the desired path.

Similar repulsive fields have been previously used as a means of preventing inter-robot and robot-obstacle collisions in static, multi-robot pattern generation scenarios [11, 12]. However these studies are focused on mobile ground robots, which do not have the forward velocity constraint imposed on fixed-wing UAVs. The effect of the OVF is restricted to a desired radius by a decay function. Several investigations of decay functions for VF obstacles have been explored with varying results [1–7]. The main choice for a guidance system designer is to select a function that allows the obstacles field strength to be high near the center and edge and then fall off once the edge threshold has been exceeded. This decay function scales the OVF magnitude based on the radial distance from a vehicle to the center of the OVF. An example OVF is shown in Fig. 3. Currently, it is unclear how the choice of decay function affects the obstacle avoidance performance of a fixed-wing UAV. The present work sought to provide insight to this question by examining the effects of several different OVF decay functions on vehicle performance.

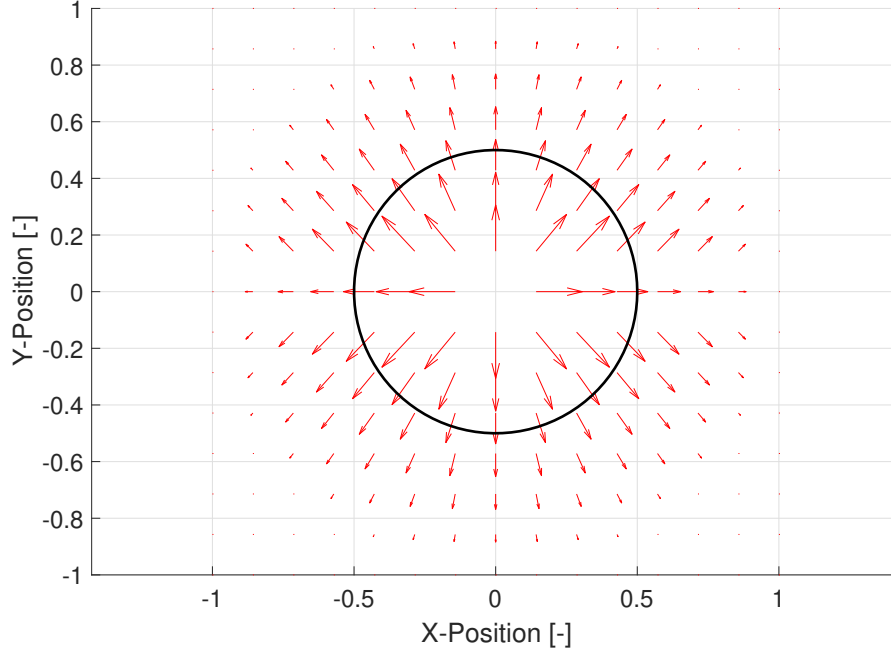


Fig. 3 An example obstacle vector field using a ramp decay function. The black line indicates the desired obstacle radius.

IV. Simulation Description

A. Dubins Vehicle

The UAV was modeled as a Dubins vehicle, represented in Equations 4-5. The vehicle velocity v is held constant while the vehicle heading θ is allowed to vary based on the angle dictated by the VF. The turn rate is limited by Equation 6.

$$\dot{x} = v \cos(\theta) \quad (4)$$

$$\dot{y} = v \sin(\theta) \quad (5)$$

$$\dot{\theta} \leq 20 \text{ deg/s.} \quad (6)$$

B. Scenarios

Simulations were run for several different obstacle avoidance scenarios. All three scenarios involved a simulated, VF-guided UAV following a straight line. This GVF was generated with the GVF method using two intersecting planes. The equations which give the straight line field components at a given departure angle of δ are as follows.

$$\vec{V}_{conv} = \frac{-1}{\sqrt{(\cos(\delta)x + \sin(\delta)y)^2 + z^2}} \begin{bmatrix} x \cos^2(\delta) + \cos(\delta) \sin(\delta)y \\ y \sin^2(\delta) + \cos(\delta) \sin(\delta)x \\ z \end{bmatrix} \quad (7)$$

$$\vec{V}_{circ} = \begin{bmatrix} \sin(\delta) & -\cos(\delta) & 0 \end{bmatrix}^T \quad (8)$$

$$\vec{V}_{TV} = 0 \quad (9)$$

The weights for the straight line following VFs are $G = 1$, $H = 1$, and $L = 0$.

The OVFs are generated with the GVF method using the intersection of a cylinder and a plane. These surfaces are defined by $\alpha_1 = (x - x_c)^2 + (y - y_c)^2 - r_{OVF}^2$, and $\alpha_2 = z$ respectively, where x_c and y_c are the x - and y -coordinates of the center of the OVF. The convergence term for this field is given by

$$\vec{V}_{conv} = A\vec{B} \quad (10)$$

where

$$A = \frac{-1}{\sqrt{\bar{x}^4 + \bar{y}^4 + 2\bar{x}^2\bar{y}^2 - 2r^2\bar{x}^2 - 2r^2\bar{y}^2 + r^2 + z^2}} \quad (11)$$

$$\vec{B} = \begin{bmatrix} 2\bar{x}^3 + 2\bar{x}\bar{y}^2 - 2r^2\bar{x} \\ 2\bar{y}^3 + 2\bar{x}^2\bar{y} - 2r^2\bar{y} \\ z \end{bmatrix} \quad (12)$$

and

$$\bar{x} = x - x_c \quad (13)$$

$$\bar{y} = y - y_c \quad (14)$$

The circulation term can be represented as

$$\vec{V}_{circ} = \begin{bmatrix} 2(y - y_c) \\ -2(x - x_c) \\ 0 \end{bmatrix}. \quad (15)$$

Preliminary work indicates that the inclusion of a circulation term in the OVF field improves obstacle avoidance performance. This is illustrated in Fig. 4 and Fig. 5. As such, all OVFs in the present work have component weights of $G = -1$, $H = \pm 5$, and $L = 0$.

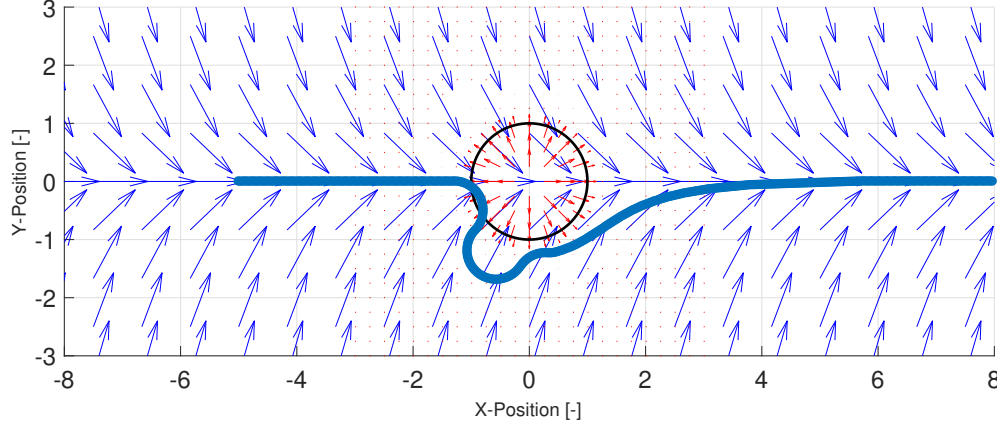


Fig. 4 Example trajectory of vehicle around a purely repulsive OVF. $G = -1$, $H = 0$, $L = 0$.

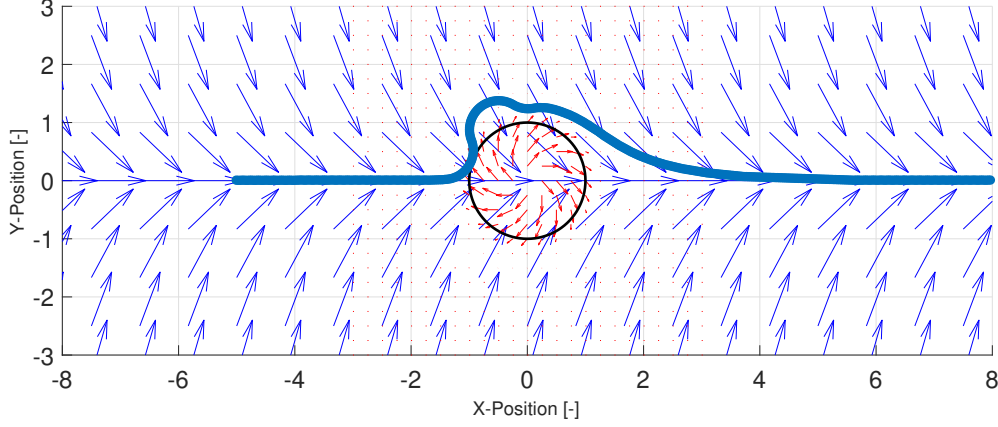


Fig. 5 Example trajectory of vehicle around an OVF with a circulation term. $G = -1$, $H = 5$, $L = 0$.

The sign of the circulation term depends on the vehicle approach angle, θ_a , and was determined using

$$H = \begin{cases} H & \text{if } \theta_a \leq 0 \\ -H & \text{if } \theta_a > 0 \end{cases} \quad (16)$$

In each scenario, qualitative behavior of the vehicle is obtained by examining its trajectory. Descriptions of each scenario are as follows:

1. Obstacle Boundary Violation

If the vehicle velocity is sufficiently large and the obstacle radius is sufficiently small, it is likely that the vehicle will violate the boundaries of the obstacle. Accordingly, the quantity of interest for boundary violation studies is the ratio of the OVF radius to the vehicle velocity, v/r . For single object obstacle avoidance, head-on collision with the OVF presents the worst possible scenario. The first simulation in the present work compares the trajectories of a vehicle subject to head-on collision for v/r values of 1, 10, 100, and 1,000. The vehicle begins five OVF radii away from the OVF center. This simulation is repeated for a given v/r using each of the different decay functions. To vary v/r , the vehicle velocity is increased while the OVF radius is held at $r = 1$.

2. Vehicle Approach Angle

Vehicles may approach OVFs from a wide variety of initial conditions. This study sought to examine the effects of different decay functions on the obstacle avoidance performance of a vehicle approaching an OVF from different angles. As with the previous simulation, this simulation only includes one OVF. $v/r = 10$ for all simulations in this study. The initial distance between the vehicle and the center of the OVF is five times the OVF radius. The initial approach angle is set to 0, 30, 60, and 90 degrees. The simulation is repeated with every decay function for each initial approach angle.

3. Multiple Obstacle Avoidance

Real vehicles utilizing VF guidance will likely need to avoid multiple obstacles in normal operation. Interacting OVFs may require decay functions that are different from the decay functions that optimize single OVF performance. The third simulation includes two OVFs that are staggered apart and two OVFs that are parallel to each other. $v/r = 10$ for this simulation as in the vehicle approach angle study. As before, the simulation was repeated using each decay function.

C. Decay Functions

A description of each of the five decay functions is given in this section along with the rationale for its selection. Each decay function depends on the nondimensionalized radius which is given by

$$\tilde{r} = \frac{r}{r_{\text{OVF}}}, \quad (17)$$

where r is the radial distance from the center of the OVF to the vehicle and r_{OVF} is the desired radius of the OVF. The peak magnitudes of the functions bounded at $\tilde{r} = 0$ are normalized to $P(\tilde{r}) = 1$. R_A is the percentage of the OVF radius at which the field strength is A . In this initial study, $R_A = 1$ and $A = 0.5$. That is, the field strength at the defined OVF radius is 50%.

1. Gaussian

The Gaussian function is found in probability as the ubiquitous normal distribution and in physics as the Green's function for the diffusion and heat equations. Its magnitude is one at the center of the OVF and asymptotically approaches 0 as the radius goes to infinity.

$$P(\tilde{r}) = A \left(\frac{\tilde{r}}{R_A} \right)^2 \quad (18)$$

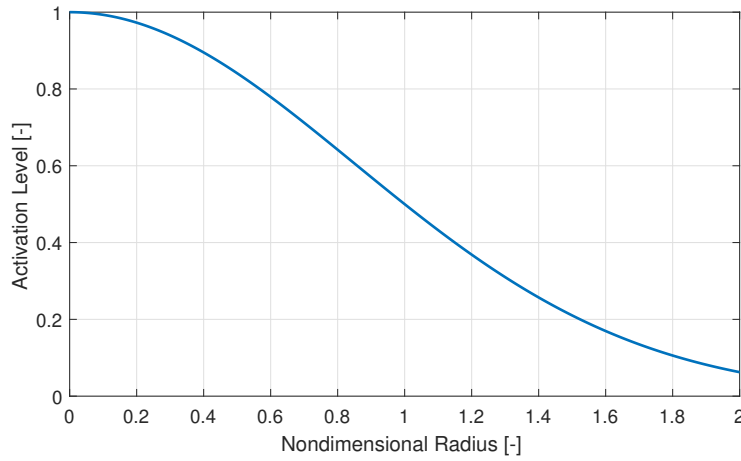


Fig. 6 Gaussian decay function

2. Hyperbolic Tangent

The hyperbolic tangent function smoothly transitions from a magnitude of one at the center of the OVF and asymptotically approaches zero as the radius goes to infinity.

$$P(\tilde{r}) = A(\tanh(2\pi(R_A - \tilde{r}))). \quad (19)$$

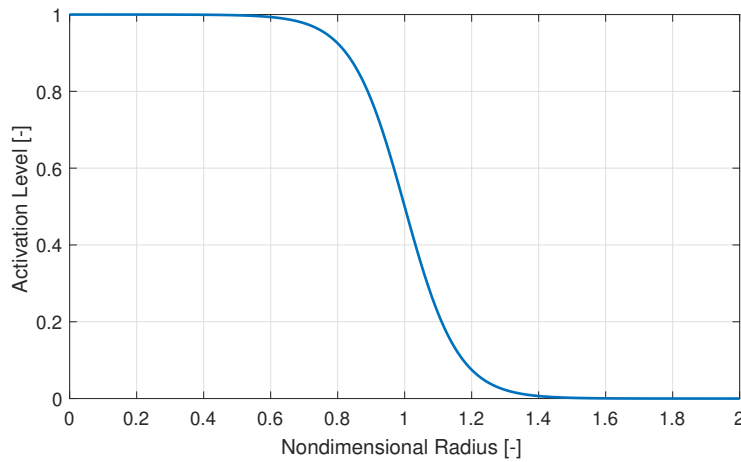


Fig. 7 Hyperbolic tangent decay function

3. Ramp

The ramp function also appears as a neural network activation function. Like the hyperbolic tangent function, it varies from a magnitude of one at the center to zero at a radius greater than the OVF radius. The transition to a magnitude of 0 is discontinuous.

$$P(\tilde{r}) = \begin{cases} \left(\frac{A-1}{R_A} \right) \tilde{r} + 1, & \text{if } 0 \leq \tilde{r} \leq \frac{-R_A}{A-1} \\ 0, & \text{otherwise} \end{cases} \quad (20)$$

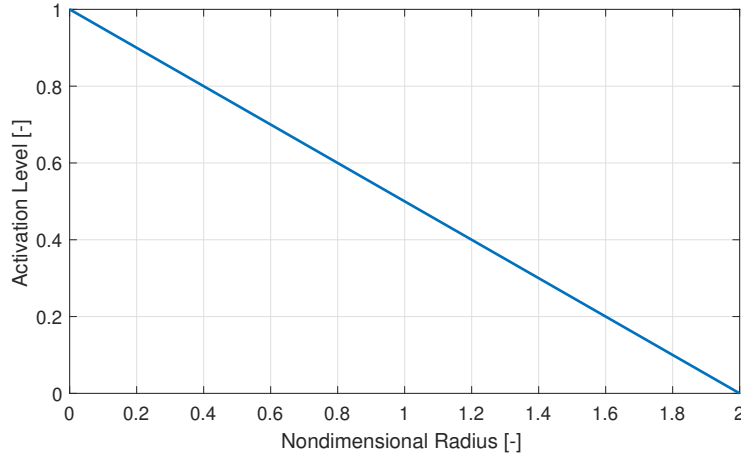


Fig. 8 Ramp decay function

4. Inverse Square

Inverse square functions frequently appear in nature, most notably as the decay function for the electric field of a point charge or the gravitational field of a point mass. Its magnitude becomes unbounded at the center of the OVF while it asymptotically approaches zero as the radius goes to infinity.

$$P(\tilde{r}) = A \left(\frac{R_A}{\tilde{r}} \right)^2 \quad (21)$$

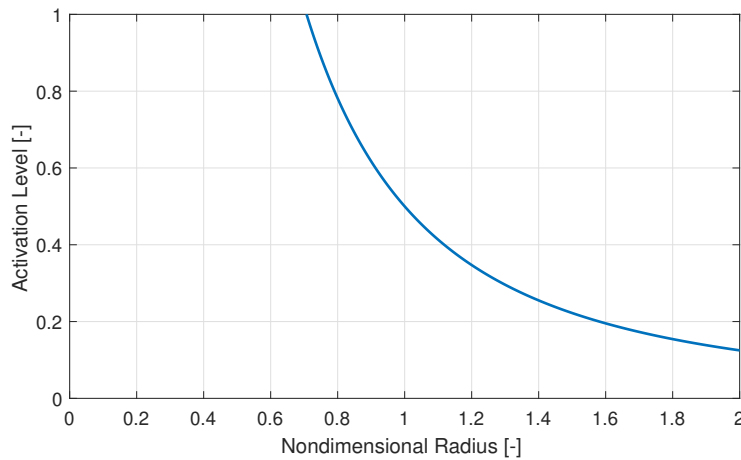


Fig. 9 Inverse square decay function

5. Inverse Exponential

Exponential decay also occurs widely in nature as a solution to certain differential equations. Its value at the center of the OVF is one and it asymptotically approaches zero as the radius goes to infinity.

$$P(\tilde{r}) = \left(\frac{1}{A}\right)^{-\frac{\tilde{r}}{R_A}} \quad (22)$$

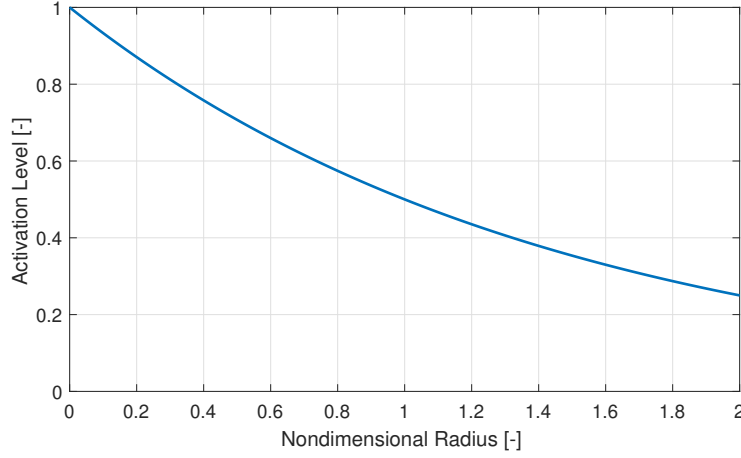


Fig. 10 Inverse exponential decay function

V. Results

A. Obstacle Boundary Violation

The results of the obstacle boundary violation study are given in Figs. 11 - 14. Three classes of behavior in the trajectory of the simulated vehicle are observed for $v/r = 1$ and $v/r = 10$. For the inverse exponential and inverse square functions, the vehicle is immediately diverted away from the obstacle. For the Gaussian and linear functions, the vehicle becomes trapped in a circular holding pattern. The hyperbolic tangent function is the only function where the vehicle is able to successfully traverse around the obstacle. At $v/r = 10$, the trajectory of the vehicle around the obstacle with the hyperbolic decay function includes a small bump. This is an artifact of the interplay between the guidance field, obstacle field, and restricted vehicle turn rate. As expected, when v/r increases, the vehicle is not able to react quickly enough to avoid the obstacle. At $v/r = 100$, the inverse square and inverse exponential functions cause the vehicle to deflect away from the obstacle.

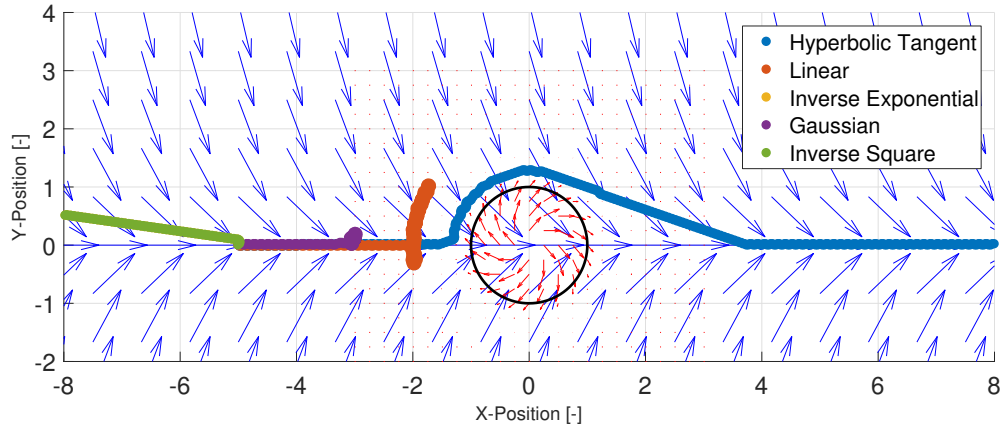


Fig. 11 Result from the obstacle boundary violation simulation for $v/r = 1$

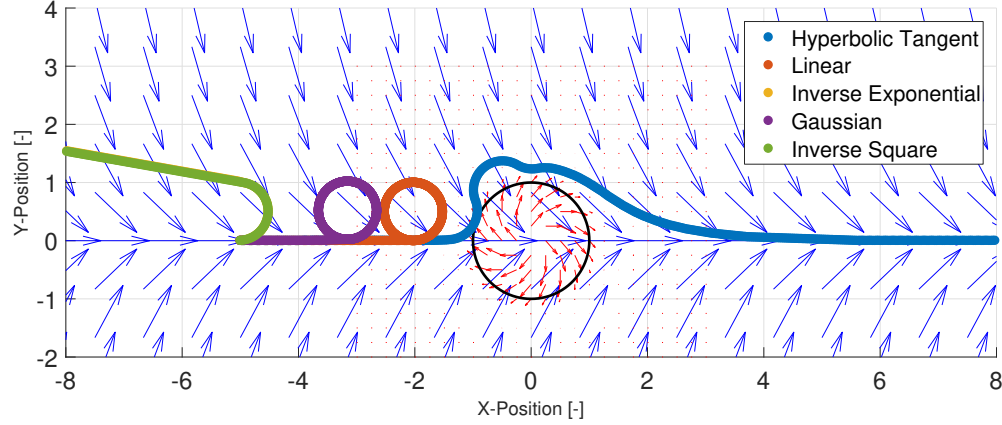


Fig. 12 Result from the obstacle boundary violation simulation for $v/r = 10$

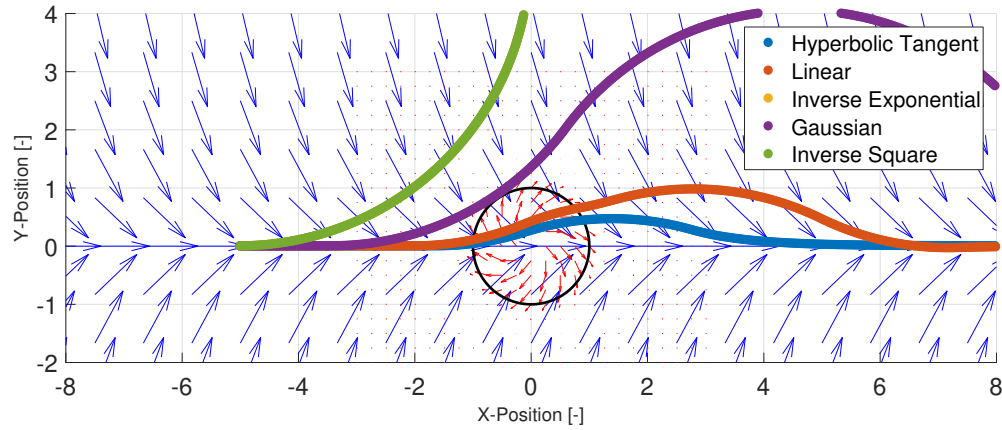


Fig. 13 Result from the obstacle boundary violation simulation for $v/r = 100$

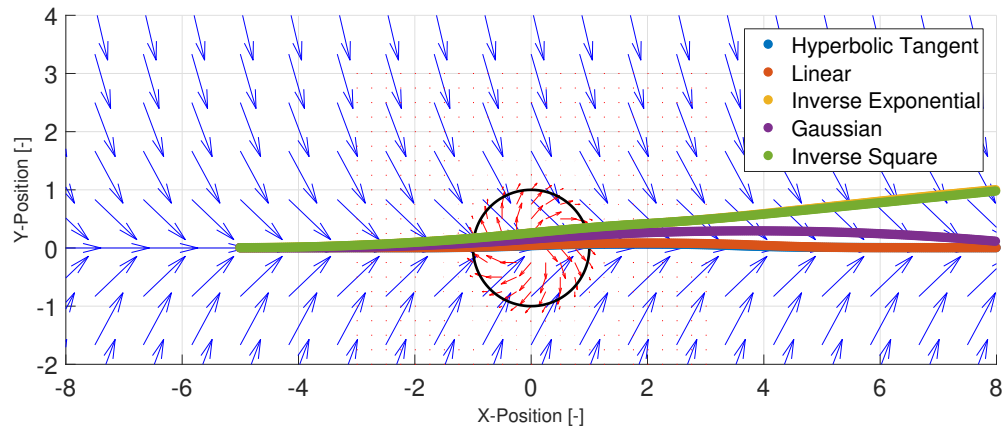


Fig. 14 Result from the obstacle boundary violation simulation for $v/r = 1000$

The Gaussian function allows for the vehicle to travel around the object, though it takes a significant amount of time for the vehicle to return to its nominal trajectory. The hyperbolic tangent and linear decay functions allow the vehicle to violate the obstacle boundaries. At $v/r = 1000$, the vehicle violates the obstacle boundary regardless of the choice of decay function. The inverse square, inverse exponential, and Gaussian functions cause a small deviation away from the

nominal trajectory after the obstacle. The hyperbolic tangent and linear functions have minimal effect on the trajectory of the vehicle.

B. Vehicle Approach Angle

For every initial approach angle, the inverse exponential and inverse square decay functions cause the vehicle to be immediately diverted away from the obstacle. For initial approach angles of 0 deg and 30 deg, the hyperbolic tangent function is the only decay function which allows the vehicle to navigate around the obstacle. As with the v/r study, for these approach angles, the linear and Gaussian decay functions cause the vehicle to remain in a holding pattern. For an initial approach angle of 60 deg, the linear and hyperbolic tangent functions allow the vehicle to traverse around the obstacle. The hyperbolic tangent function causes a smaller deviation of the vehicle from the nominal path when compared to the Gaussian linear. At an initial approach angle of 90 deg, the hyperbolic tangent, Gaussian, and linear functions all allow the vehicle to get around the obstacle. The linear and hyperbolic tangent functions result in identical trajectories, while the Gaussian function causes the vehicle to take a path with a larger tracking error. The results of the vehicle approach angle study are shown in Figs. 15, 16, and 17.

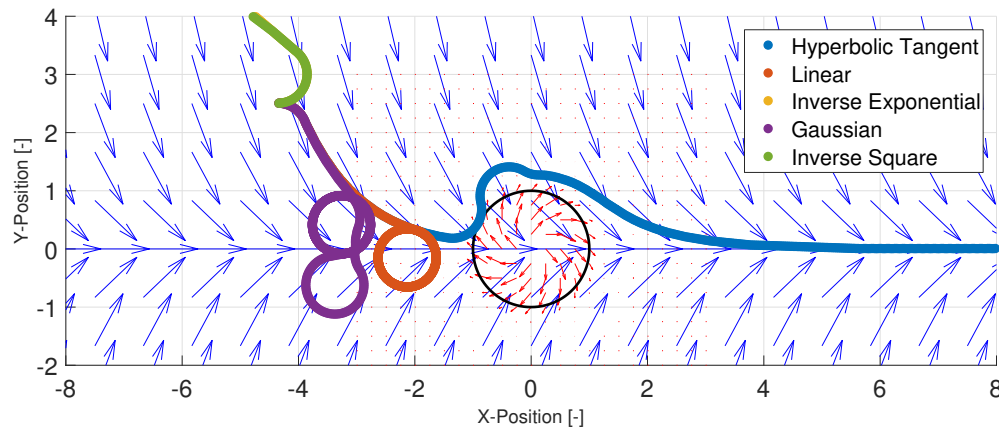


Fig. 15 Result from the vehicle approach angle simulation for $\theta_0 = 30$

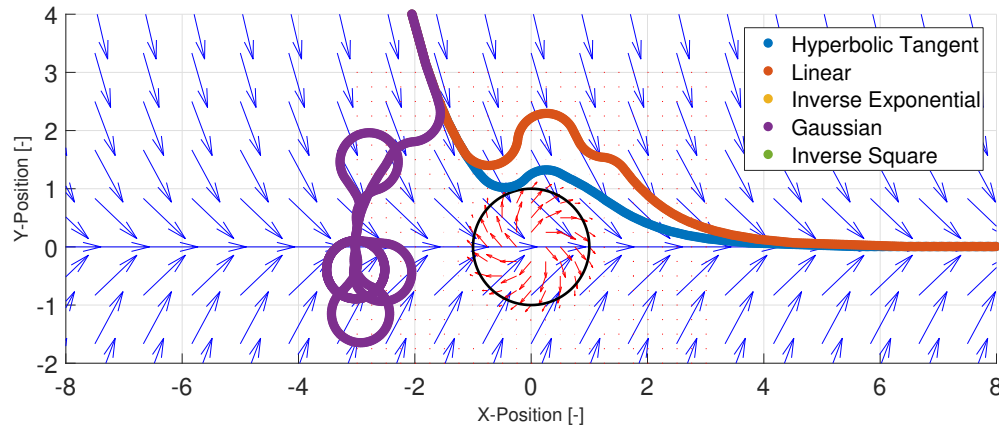


Fig. 16 Result from the vehicle approach angle simulation for $\theta_0 = 60$

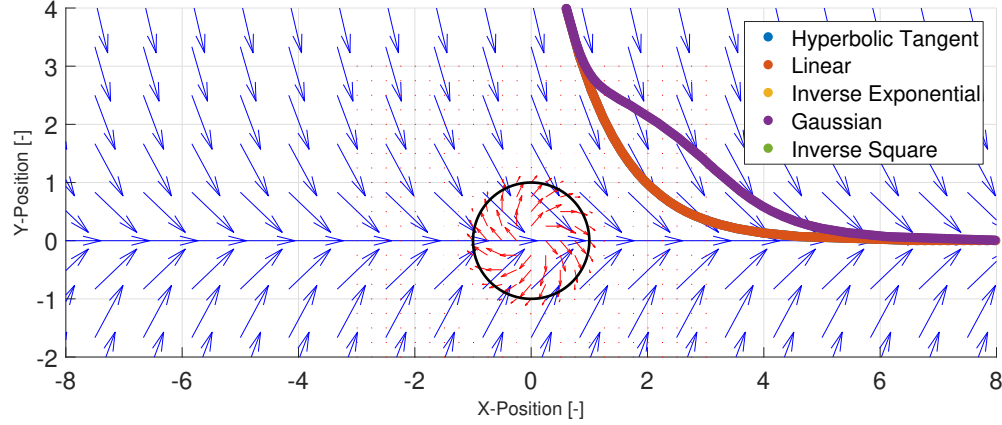


Fig. 17 Result from the vehicle approach angle simulation for $\theta_0 = 90$

C. Multiple Obstacle Avoidance

The results of the multiple obstacle avoidance study are presented in Fig. 18. As with the previous two studies, the hyperbolic tangent function allows the vehicles to navigate the multiple obstacles, the linear and Gaussian functions cause the vehicle to maintain a holding pattern, and the inverse square and inverse exponential functions cause the vehicle to divert away from the obstacles.

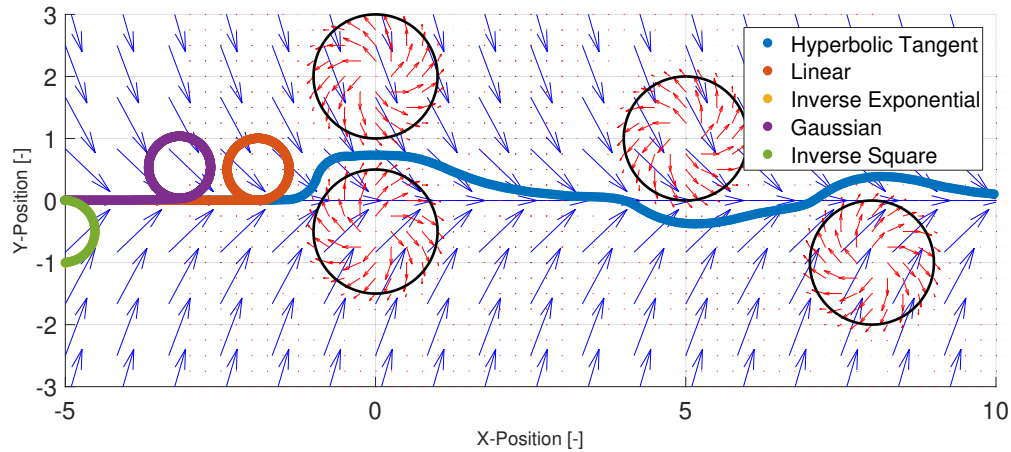


Fig. 18 Result from the multiple obstacle avoidance scenario

D. Modified Decay Functions

Head-on collision simulations of $v/r = 10$ were re-run for each decay function. For this simulation, the decay functions were modified by setting $P(\tilde{r}) = 0$ for $\tilde{r} \geq 1.4$. This simulation was conducted to determine if the effectiveness of the hyperbolic tangent function was due to its sharp drop-off in magnitude, or if it was due to something else, perhaps the curvature of the function. The results of this are given below in Fig. 19. As can be seen, the vehicle successfully navigates around the obstacle for every modified decay function. Interestingly, the vehicle trajectories for the modified linear, Gaussian, inverse exponential, and inverse square functions are identical.

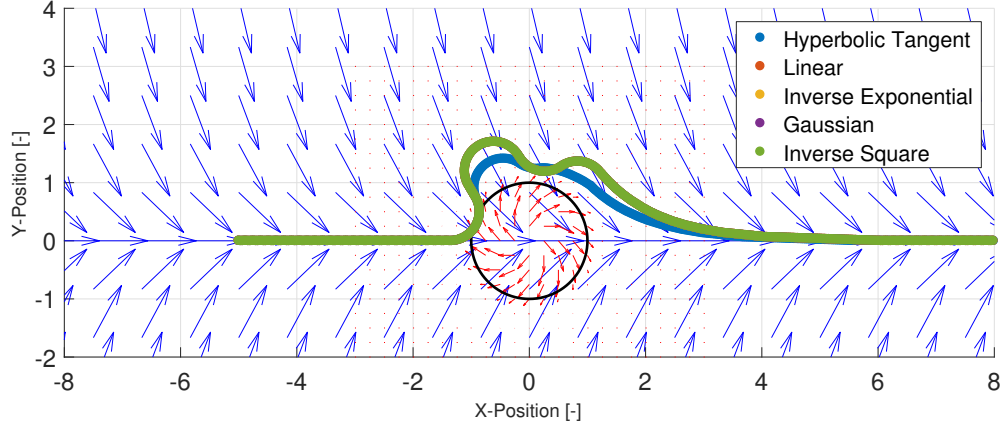


Fig. 19 Results of a head-on collision simulation using modified decay functions

E. Experimental Results

Real-time experiments were ran to verify simulation results for two obstacles using the hyperbolic tangent decay and linear decay. Linear and hyperbolic tangent decay were chosen as they present the most successful overall simulation results. Experiments were ran using a CrazyFlie 2.0 (CF) quad-rotor with a 20 deg/s turn rate constraint to emulate a fixed wing UAV. Guidance commands were determined from GVF solutions while varying the decay function used to create the obstacle field. The CF vehicle was chosen due to test space limitations and Vertical Takeoff Or Landing (VTOL) capabilities. The CF is equipped with an Inertial Measurement Unit (IMU) for vehicle stability but no local position control for guidance. Feedback for the CF's position was obtained using eight Vicon Vero motion capture cameras interfaced with the Vicon Tracker software. Position data was taken from the Vicon system and used as feedback for a Python based controller and guidance interface. Commands for roll, pitch, yaw rate, and throttle position were then sent to the CF ground station software (CfClient) from the Python interface. CfClient then sent commands to the vehicle wirelessly. The control and guidance system included GVF algorithms and Proportional-Integral-Derivative (PID) controllers for x position, y position, z position, and yaw control. Set points using a virtual carrot chasing point which represent a point in the heading direction with respect to the CF's body were used for guidance [13]. Navigation and control systems operated at 100 Hz while the GVF guidance was set to run at 10 Hz to avoid over stimulation of guidance. Fig. 20 presents an overview of the experimental setup.

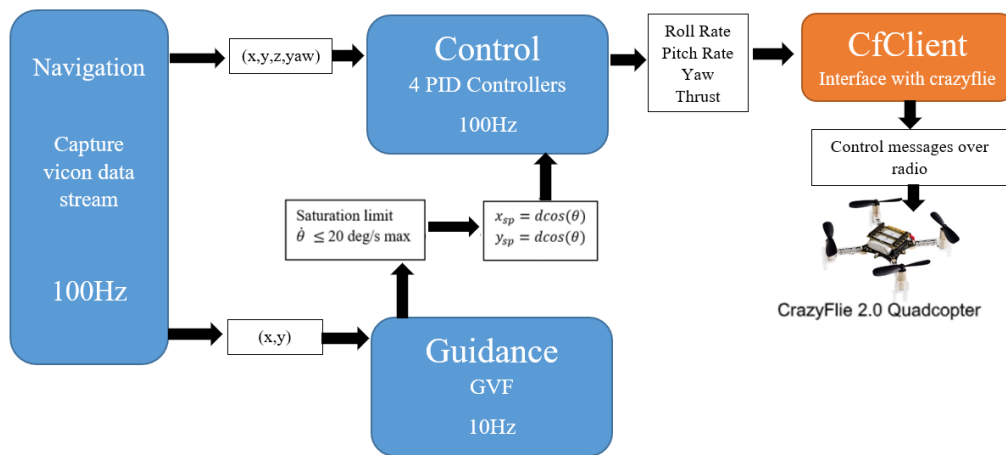


Fig. 20 Guidance, navigation, and control framework for crazyflie 2.0 quadcopter

The experimental setup was used to determine the performance of the linear and hyperbolic tangent decay functions when applied to obstacles centered at $(x = -0.75, y = 0.15)$ m and $(x = 0.25, y = -0.15)$ m both with a radius of 0.25 m. The locations and radii for the obstacles were chosen based on the field of view for the Vicon system in the space

provided. The UAV path was set as a straight line, where $\delta = 0$. All simulation and experiments were ran at a mean UAV velocity of 0.12 m/s resulting in $v/r = 0.48$ starting at $(x = -1.5, y = 0)$. Altitude for the UAV was set at 1.4 m while the vehicle approach is initially in line with the path. Simulation and experimental results for both cases present the same trend when compared to their respective decay. All methods avoid the two obstacles with the hyperbolic tangent experiment getting close to the second obstacle from the left. Both decay methods prove successful but the hyperbolic tangent decay has less deviation from the planned path than linear as shown in Fig. 21.

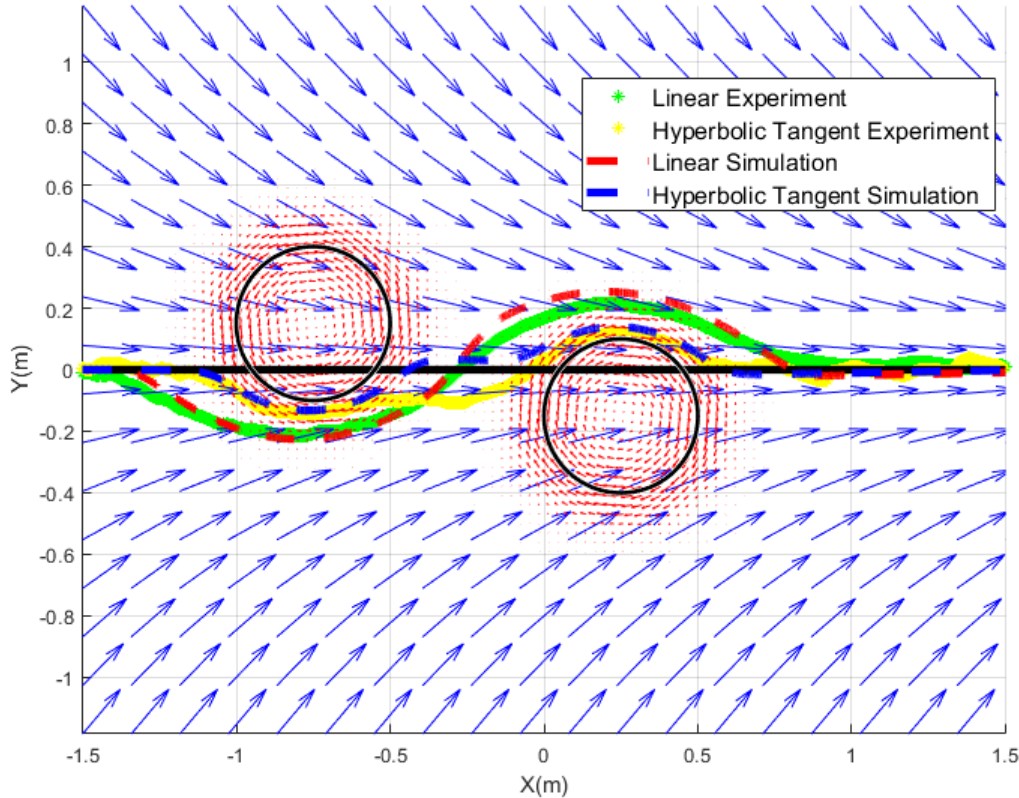


Fig. 21 Experimental and simulation results for two object scenario

VI. Conclusion

In the present work, the impact of decay functions on obstacle avoidance performance was studied in simulation and verified experimentally. Hyperbolic tangent, linear, Gaussian, inverse exponential, and inverse square decay functions were studied in multiple simulation environments. All simulations were modeled using Dubins vehicles with GVF guidance with obstacles along a straight path. Simulation scenarios with varying v/r , vehicle approach angle, and a case with multiple objects were ran. Each of these variables were varied while all others were held constant to compare results on even ground. Experiments on a two obstacle scenario using hyperbolic tangent and linear decay functions were ran to verify real time flight performance. A CF UAV with turn rate constraints to emulate fixed wing dynamics was used in a Vicon environment to gather experimental data. Flight performance data was compared to simulations results and proved to be successful, verifying the real-time capabilities of the presented work. Three observations regarding the impact of decay functions on obstacle avoidance performance may be made from the results of the present work. First, an effective decay function must rapidly approach zero once the radial distance exceeds the OVF radius. This is supported by the success of the hyperbolic tangent function in the majority of the scenarios. The hyperbolic tangent function drops off to zero at a much faster rate than the inverse exponential, inverse square, and Gaussian functions. The linear function also shows some measure of success and also approaches zero much faster than the three

poorly performing decay functions. The present work indicates that the hyperbolic tangent function, and other functions that quickly drop off to zero once outside the OVF boundary, are superior for $v/r \leq 10$. Second, when including a circulation term within an OVF, obstacle avoidance performance is improved. Third, these obstacle fields are only effective for a range of certain v/r values. The vehicle will violate the boundary of an OVF using a hyperbolic tangent decay function for $v/r > 10$. The longer decay distances of the linear and Gaussian functions allow them to be used for obstacle avoidance at $20 \leq v/r \leq 30$ and $30 \leq v/r \leq 40$ respectively. However this is somewhat analogous to simply changing the obstacle size and reducing v/r . The impacts of this work include the first study of decay functions applied to OVFs in GFV based UAV guidance and the identification of the best performing decay method for the given scenarios. For the tested decay functions, the hyperbolic tangent decay function proved to outperform all others for a $v/r < 10$.

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